

Lab Assignment 4

06/02/2026

Total Marks: 20

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- Write the query, execute it, paste the screenshots, and make a PDF. Paste the query text too. You need to show a screenshot to demonstrate the correctness of your query. If you have made some changes that are not visible in the direct output, display the whole table again.
 - The file name should be **<roll no>_<name>_assignment4.pdf**
 - Write the questions before the answers.
 - Write a small justification for your solution.
 - You can use **PGAdmin** for this lab, and make sure to submit the **.sql** in Moodle
 - Complete the exercise before 4.45 PM, then submit the files on Moodle.
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1. Write a query that compares film_id with the string '10', with and without using typecasting. (0.5 + 0.5 marks)
2. List all rentals that occurred only on the date 25-05-2005 (1 mark)
3. Find all the movies that were never rented. (Display film_id, title). (1.5 mark)
4. Display the details (customer_id) of each customer and the total number of films rented by that customer. (1.5 marks)
5. Find the customer ID, customer full name, staff ID, staff full name, and payment ID, where the customer's last name matches that of the staff member to whom they made a payment, but the first names do not match. (2 marks)
6. Find all distinct pairs of actors whose first names do not match, using only an inner join (no where clause). Make sure there are no repeated pairs, i.e., (1,2) and (2,1). (2 marks)
7. For each customer, find the corresponding actors' details whose films they have rented. (Use Natural join, and column renaming is not allowed) (1.5 marks)

8. A. Find the count of movies for which no rental exists with a rental_date after 25-05-2005. (1.5 marks)

B. From the above result, ensure that those movies were returned before 25-06-2005. (1 mark)
9. For all possible distinct pairs of category and language, count the number of movies. (All possible pairs of categories and language must be displayed.) (2 marks)
10. For each store, generate a staff-customer interaction analysis report showing all possible combinations of staff members and customers in the database, regardless of whether they have actually interacted. Make sure the customer has made at least one rental on or after 2006-01-26. The report should have the Store ID, Staff ID and name, Customer ID and name, and a field showing whether this staff has handled the rental by that customer. (3 marks)
11. List all customers who have rented movies, but have never rented a movie from the 'Horror' category. Use a nested query and a LEFT JOIN.(2 marks)

